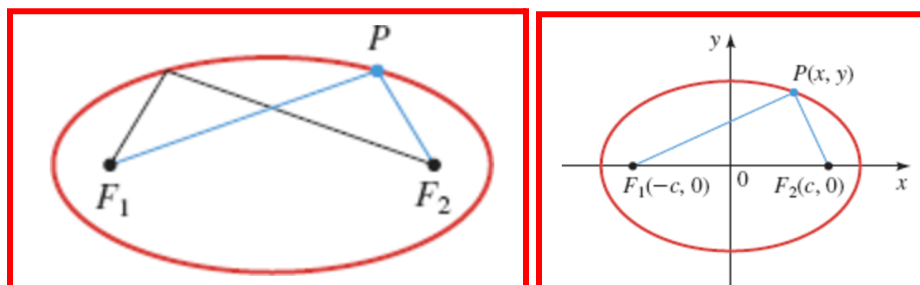


An **ellipse** is an oval curve that looks like an **elongated circle**.

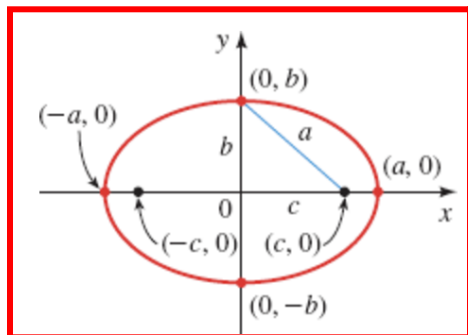
Geometric Definition of an Ellipse

An ellipse is the set of all points in the plane the **sum of whose distances from two fixed points** F_1 and F_2 is a **constant**. These two fixed points are the **foci** (plural of focus) of the ellipse.



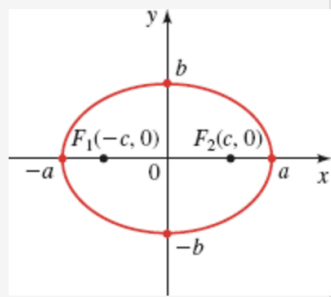
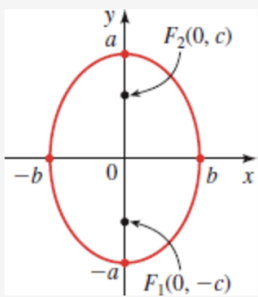
The standard equation of an ellipse with foci on the x -axis and center at the origin is

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$). If we let $y = 0$ then $x = \pm a$. The ellipse crosses the x -axis at $(a, 0)$ and $(-a, 0)$, as shown below. These points are called the **vertices** of the ellipse, and the segment that joins them is called the **major axis**. Its length is $2a$. If we let $x = 0$ then $y = \pm b$. The ellipse crosses the y -axis at $(0, b)$ and $(0, -b)$. The segment that joins these points is called the **minor axis**, and it has length $2b$. If the foci of the ellipse are placed on the y -axis at $(0, \pm c)$ rather than on the x -axis, then the roles of x and y are reversed in the preceding discussion, and we get a **vertical ellipse**.



Ellipse with Center at the Origin

The graph of each of the following equations is an ellipse with center at the origin and having the given properties.

EQUATION	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $a > b > 0$	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ $a > b > 0$
VERTICES	$(\pm a, 0)$	$(0, \pm a)$
MAJOR AXIS	Horizontal, length $2a$	Vertical, length $2a$
MINOR AXIS	Vertical, length $2b$	Horizontal, length $2b$
FOCI	$(\pm c, 0)$, $c^2 = a^2 - b^2$	$(0, \pm c)$, $c^2 = a^2 - b^2$
GRAPH		

In the standard equation for an ellipse, a^2 is the *larger denominator*, and b^2 is the *smaller*. To find c^2 , we subtract: **larger denominator minus smaller denominator**.

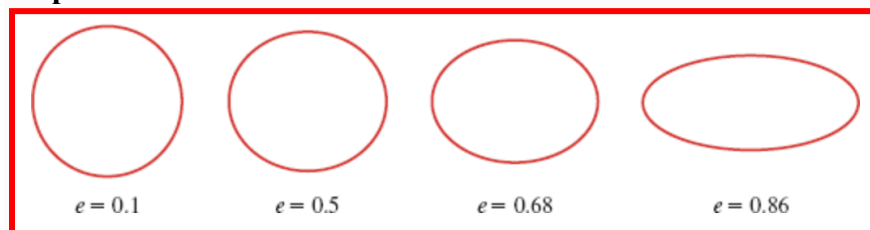
Definition of Eccentricity

For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ or $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ (with $a > b > 0$), the eccentricity e is the number $e = \frac{c}{a}$ where $c = \sqrt{a^2 - b^2}$. The eccentricity of every ellipse satisfies $0 < e < 1$.

If $a = b$ in the equation of an ellipse, then $\frac{x^2}{a^2} + \frac{y^2}{a^2} = 1$ so $x^2 + y^2 = a^2$. In this case the “ellipse” is a circle with radius a and eccentricity 0 .

Thus if e is close to 1, then c is almost equal to a , and the ellipse is elongated in shape, but if e is close to 0, then the ellipse is close to a circle in shape. The eccentricity is a measure of how “stretched” the ellipse is.

Ellipses with various eccentricities



Example 1: An equation of an ellipse is given.

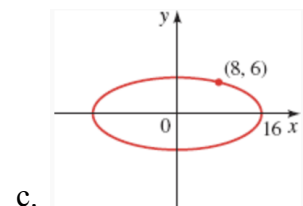
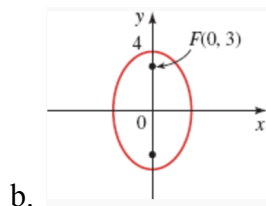
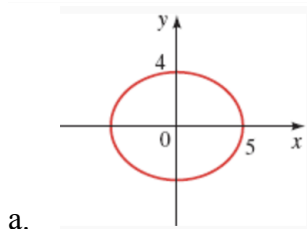
- (a) Find the vertices, foci, and eccentricity of the ellipse.
- (b) Determine the lengths of the major and minor axes.
- (c) Sketch a graph of the ellipse.

i. $\frac{x^2}{9} + \frac{y^2}{64} = 1$

ii. $3x^2 + 4y^2 = 12$

iii. $x^2 = 4 - 2y^2$

Example 2: Find the standard equation for the ellipse whose graph is shown.



Example 3: Find the standard equation for the ellipse that satisfies the given conditions.

- a. Foci: $(\pm 4, 0)$, vertices: $(\pm 5, 0)$

b. Foci: $(0, \pm 2)$, length of minor axis: 6

c. Length of major axis: 4, length of minor axis: 2, foci on y -axis

d. Endpoints of minor axis: $(0, \pm 3)$, distance between foci: 8

e. Length of minor axis: 10, foci on y -axis, ellipse passes through the point $(\sqrt{5}, \sqrt{40})$.

f. Eccentricity: $\frac{\sqrt{5}}{3}$, foci on x -axis, length of major axis: 12

Example 4: Find the intersection points of the pair of ellipses. Sketch the graphs of each pair of equations on the same coordinate axes, and label the points of intersection.

a.
$$\begin{cases} 4x^2 + y^2 = 4 \\ 4x^2 + 9y^2 = 36 \end{cases}$$

b.
$$\begin{cases} \frac{x^2}{16} + \frac{y^2}{9} = 1 \\ \frac{x^2}{9} + \frac{y^2}{16} = 1 \end{cases}$$